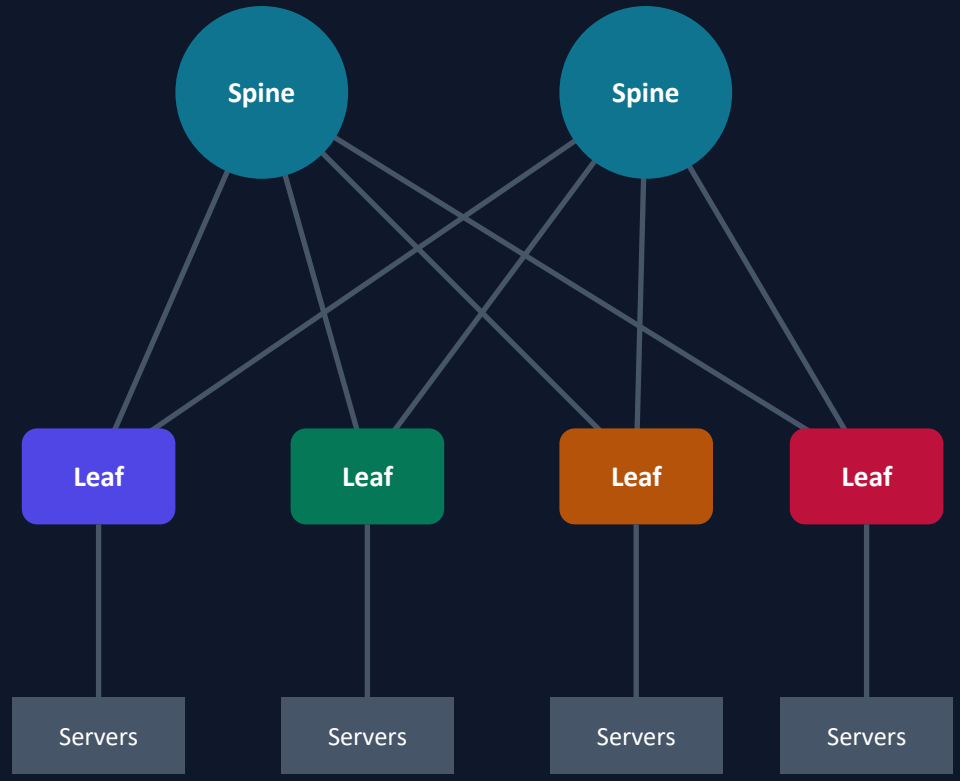
UNIT-4A

Data Centre Networks: Environment, Operations & Transport

From classical 3-tier designs to leaf-spine fabrics, and from availability and multi-tenancy to TCP in data centres



Cloud foundations: Essentials of Cloud Computing, K. Chandrasekaran

Learning Objectives & Session Plan

Module 1
Data Centre Environment · 25 min

- 3-tier architecture and its limits
- CEDSS, SDN and NFV
- Layout, cabling, storage, fabrics
- Cooling, PUE, traffic, leaf-spine

Module 2
Operational Networking Issues · 20 min

- Availability: mLAG, LACP, BFD
- Oversubscription
- Jitter and packet reordering
- VXLAN and micro-segmentation

Module 3
Transport Layer Issues · 25 min

- Why TCP struggles in DCNs
- Incast and outcast
- Queue build-up, buffer pressure
- Pseudocongestion, ECN and DCTCP

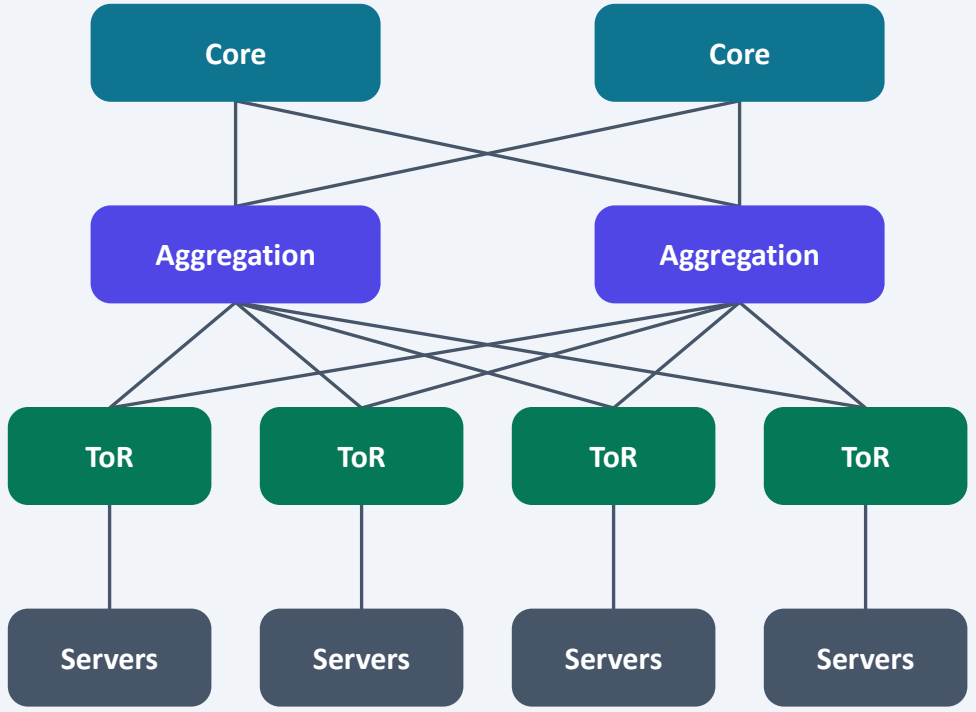
Learning outcome: after this session you should be able to explain how a cloud data centre network is built, why its design differs from classical networks, what operational problems arise, and how TCP is adapted to microsecond, shallow-buffer environments.

Classical Data Centre: The 3-Tier Tree Topology

A classical data centre network is built as a tree with three layers. The core layer is the backbone that links the data centre to external WANs and the Internet, using highly redundant layer-3 switches. The aggregation (distribution) layer joins many access switches, enforces firewall rules, manages VLANs and performs load balancing. The access layer holds the Top-of-Rack (ToR) switches that attach directly to physical servers.

Key points

- Core: L3 backbone to WAN and Internet
- Aggregation: firewalls, VLANs, load balancing
- Access: ToR switches wired to servers
- Redundant switches at every tier



Three-tier tree: core, aggregation and access layers

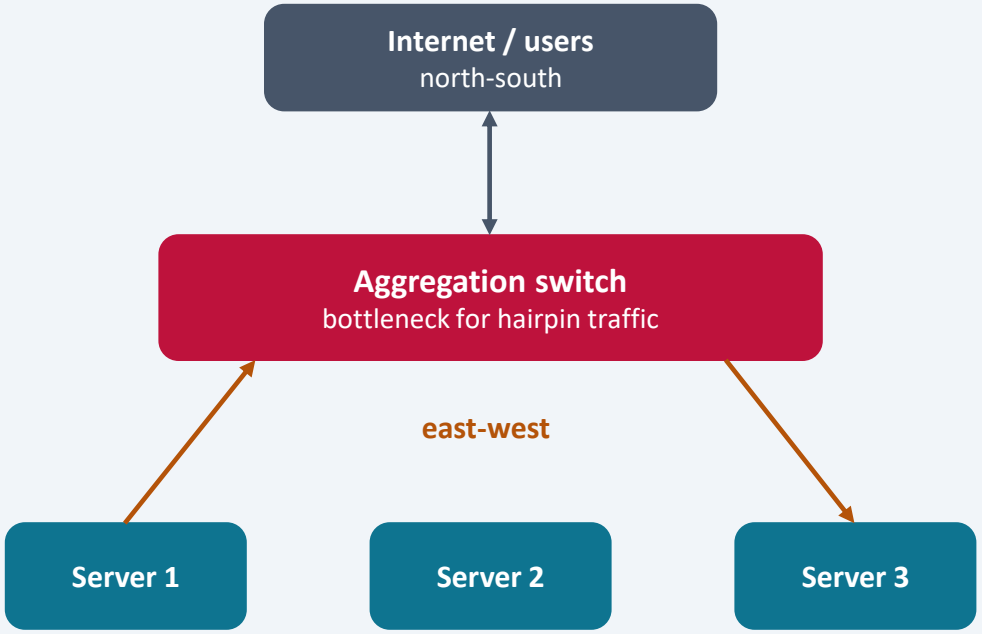
Example: A bank's web portal: a customer request enters through the core, is checked by firewalls at the aggregation layer, and reaches a web server through its rack's ToR switch; the reply retraces the same path.

Limitation: North-South Design vs East-West Demand

Classical networks were designed for north-south traffic: an external user's request travels down the tree to one server and the reply goes back up. Cloud applications behave differently. Services call each other, databases replicate and big-data jobs shuffle data, so most traffic flows east-west between servers. When this traffic must climb to the aggregation layer and come back down, the aggregation switches and their uplinks become a bottleneck.

Key points

- Designed for user-to-server (north-south) flows
- Cloud needs server-to-server (east-west) flows
- East-west traffic hairpins via aggregation
- Aggregation uplinks saturate under load



Server 1 to Server 3 must climb to aggregation and return

East-west traffic hairpins through the aggregation layer

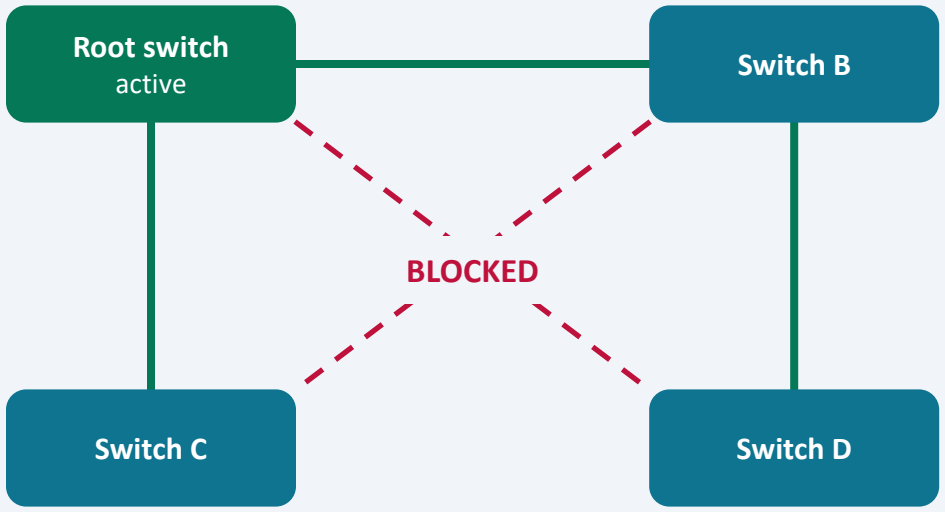
Example: Two servers in different racks exchanging a 10 GB database replica must send every byte up to an aggregation switch and back down, competing with all other racks that share that uplink.

Limitation: Spanning Tree Protocol (STP) Blocking

Layer-2 networks must not contain loops, or broadcast frames circulate forever. The Spanning Tree Protocol prevents loops by electing a root switch and deliberately disabling every redundant link, leaving a single active path. The links exist for resilience, but while the network is healthy they carry no traffic, so up to 50% of the installed link capacity is wasted. Classic STP also takes seconds to reconverge after a failure.

Key points

- Loops would cause broadcast storms
- STP blocks redundant links to form a tree
- Up to 50% of link capacity sits idle
- Slow reconvergence after failures



green = forwarding · red dashed = blocked by STP

STP keeps one active tree and blocks the other links

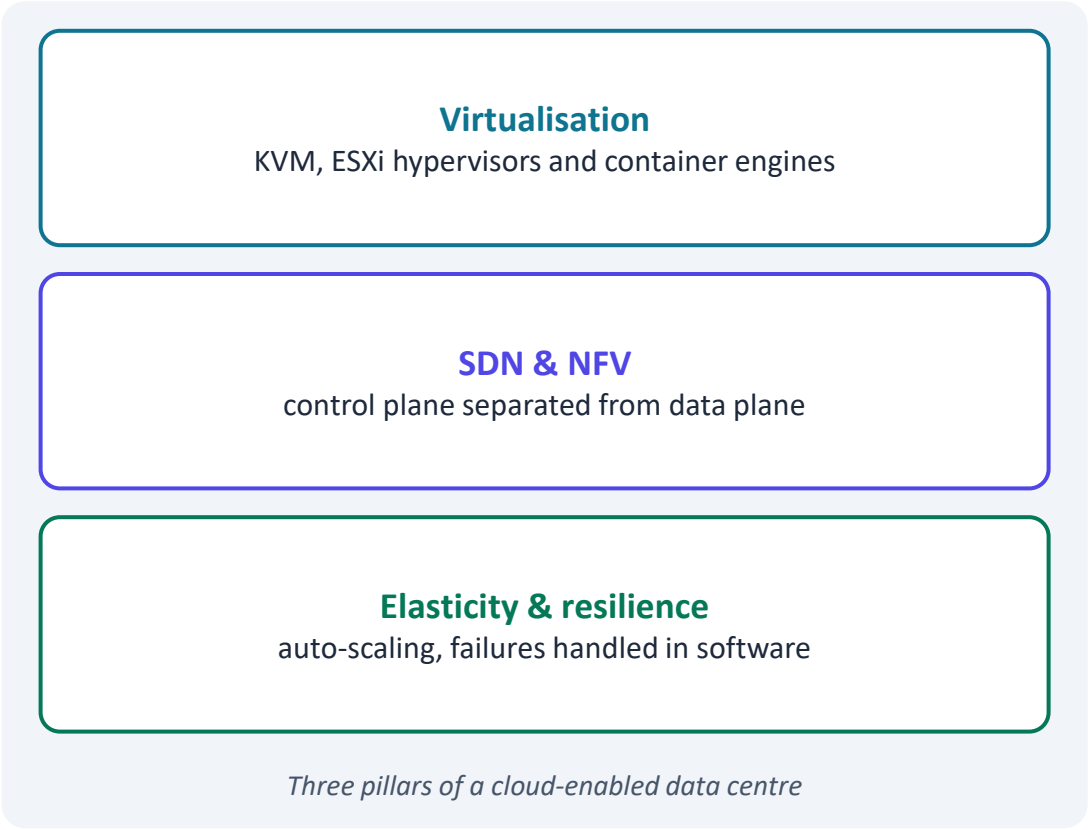
Example: A rack dual-connected to two aggregation switches uses only one uplink; the second is blocked until the first fails, so the rack never gets the double bandwidth it was cabled for.

Cloud-Enabled Data Centre Systems (CEDSS)

A cloud-enabled data centre moves from static, hardware-bound silos to software-driven, elastic pools of compute, storage and network resources. Instead of dedicating a server to one application, resources are virtualised, pooled and handed out on demand through automation. Three pillars support this shift: virtualisation, software-defined networking with network function virtualisation, and elasticity with built-in resilience.

Key points

- From hardware silos to elastic pools
- Virtualisation decouples software from hardware
- SDN and NFV make the network programmable
- Self-healing, multi-tenant design



Three pillars of a cloud-enabled data centre

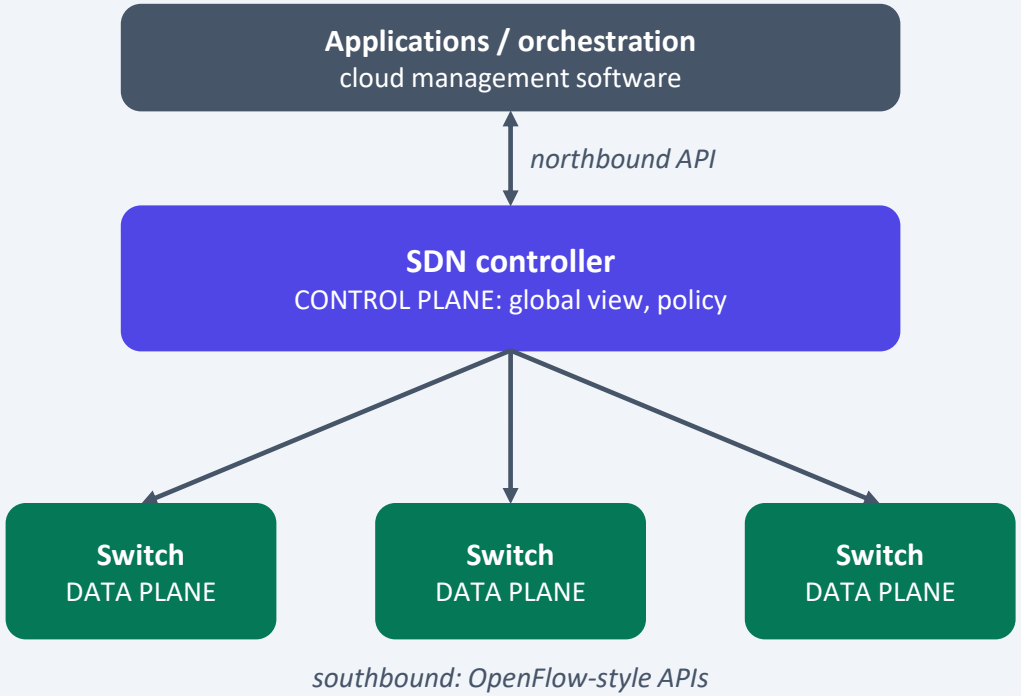
Example: On Amazon EC2 or an OpenStack private cloud a user requests a VM through an API; the platform picks a host, attaches storage and configures network isolation in seconds, with no manual wiring.

SDN & NFV: Programmable Data Centre Networks

Software-Defined Networking separates the control plane, which decides where traffic should go, from the data plane, which simply forwards packets. A logically centralised controller holds the network view and programs every switch through APIs, so thousands of switches can be provisioned like software. Network Function Virtualisation complements it by running functions such as firewalls and load balancers as software instances instead of dedicated boxes.

Key points

- Controller = control plane; switches = data plane
- Switches programmed through open APIs
- NFV runs firewalls and load balancers as software
- Faster provisioning, fewer manual errors

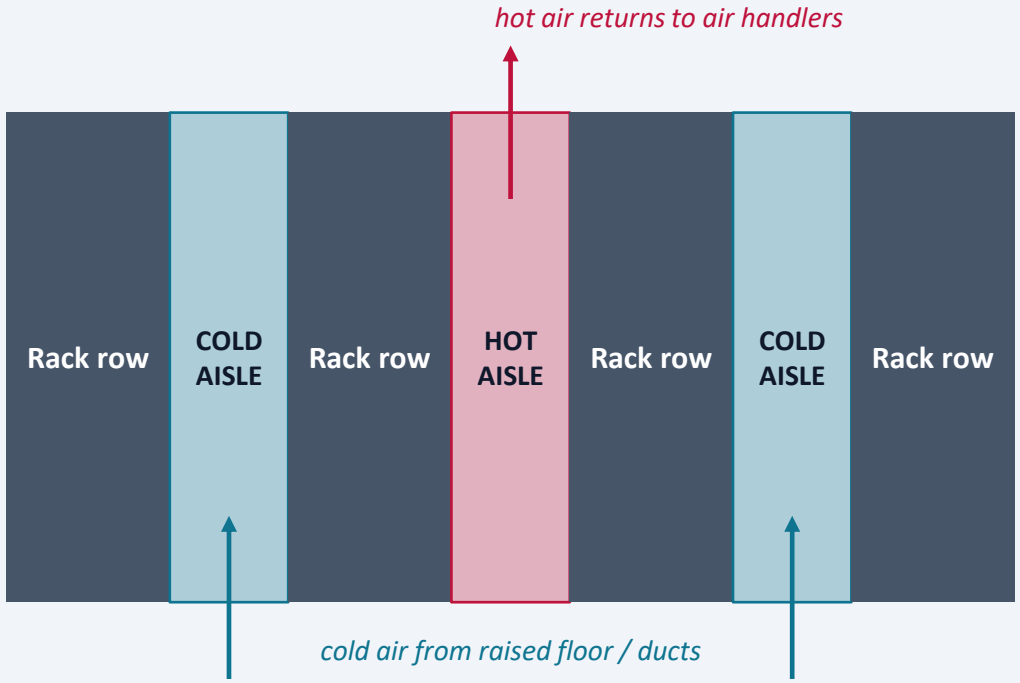


SDN: a central controller programs data-plane switches

Example: When a new tenant is created, the controller pushes the virtual network, firewall rules and load-balancer settings to all relevant switches automatically; no engineer logs in to each device.

Physical Layout: Hot Aisle / Cold Aisle Containment

Servers produce a great deal of heat, so rack placement is part of the cooling design. Racks are arranged face-to-face and back-to-back, forming alternating cold and hot aisles. Server fronts draw cold air from the cold aisle, supplied through a raised floor or ducts, and exhaust hot air into the hot aisle behind. Containing the aisles stops the two air streams mixing, and the hot exhaust returns straight to the air handlers.



Top view: racks face each other across cold aisles, back to back at hot aisles

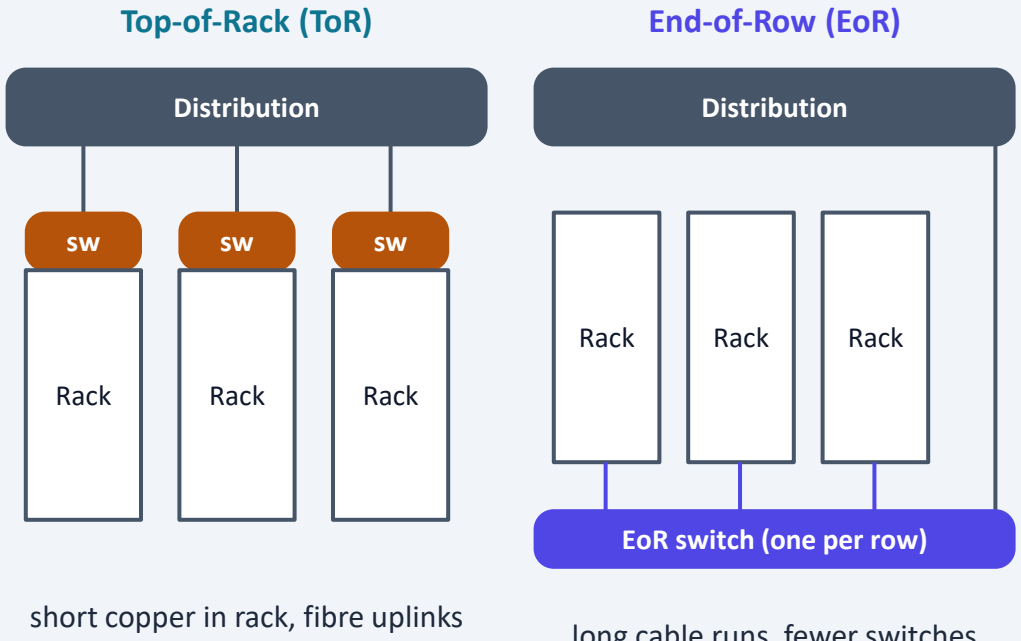
Key points

- Fronts face the cold aisle (air intake)
- Backs face the hot aisle (exhaust)
- Containment prevents hot and cold air mixing
- Hot air returns directly to air handlers

Example: In a colocation hall, doors or curtains seal the ends of the cold aisle; the same cooling units then need less fan power to keep every server inlet within its safe temperature.

Cabling Infrastructure: ToR vs End-of-Row

How switches are placed decides how servers are cabled. In a Top-of-Rack design every cabinet has its own switch; servers connect with short copper cables and only a few high-speed fibre uplinks leave the rack. In an End-of-Row design one centralised switch serves a whole row of cabinets, which needs fewer switches but long cable runs from every server back to the end of the row.



ToR places a switch in each rack; EoR centralises one switch per row

Key points

- ToR: switch in each rack, short copper cables
- ToR: fibre uplinks to the distribution point
- EoR: one switch serves a whole row
- EoR: fewer switches, long and bulky cabling

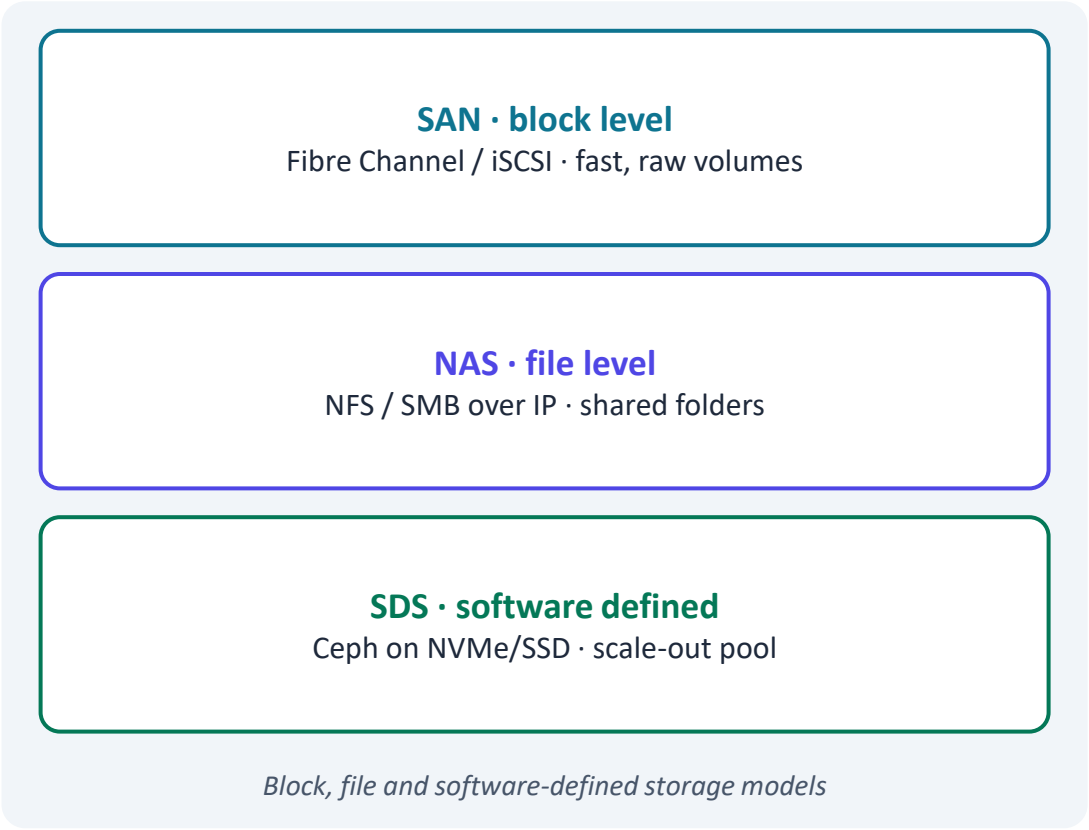
Example: Modern cloud halls favour ToR because adding a rack just means plugging in a self-contained unit; older halls with large chassis switches often use EoR to reduce the number of switches.

Storage Infrastructure: SAN, NAS and SDS

Data centres combine several storage models. A Storage Area Network (SAN) delivers fast, raw block-level storage over Fibre Channel or iSCSI, ideal for databases. Network Attached Storage (NAS) shares files over ordinary IP using NFS or SMB. Software-Defined Storage (SDS), such as Ceph, pools NVMe and SSD drives across standard servers into one distributed, scalable storage service.

Key points

- SAN: block-level over Fibre Channel or iSCSI
- NAS: file-level shares via NFS or SMB
- SDS: Ceph pools drives across commodity servers
- Choose by workload: speed, sharing, scale



Block, file and software-defined storage models

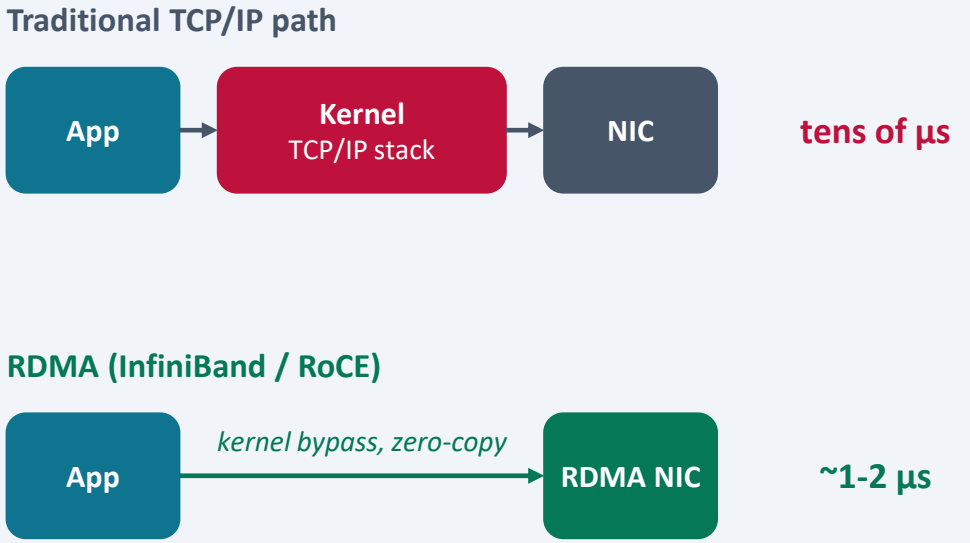
Example: A database VM uses SAN block volumes for speed, a department keeps shared documents on a NAS, and an OpenStack cloud uses Ceph to offer block and object storage from the same server pool.

Ultra-Low-Latency Fabrics: InfiniBand & RoCE

High-Performance Computing and AI training clusters exchange enormous volumes of data between servers and cannot afford the delay of the normal operating-system network stack. InfiniBand and RoCE (RDMA over Converged Ethernet) use Remote Direct Memory Access: the network card moves data straight between the memory of two servers, bypassing the kernel, so transfers complete in microseconds with very little CPU use.

Key points

- RDMA copies memory to memory over the network
- Kernel stack is bypassed (zero-copy)
- InfiniBand: dedicated, lossless fabric
- RoCE: RDMA over converged Ethernet



Kernel TCP/IP path vs RDMA kernel bypass (typical latencies)

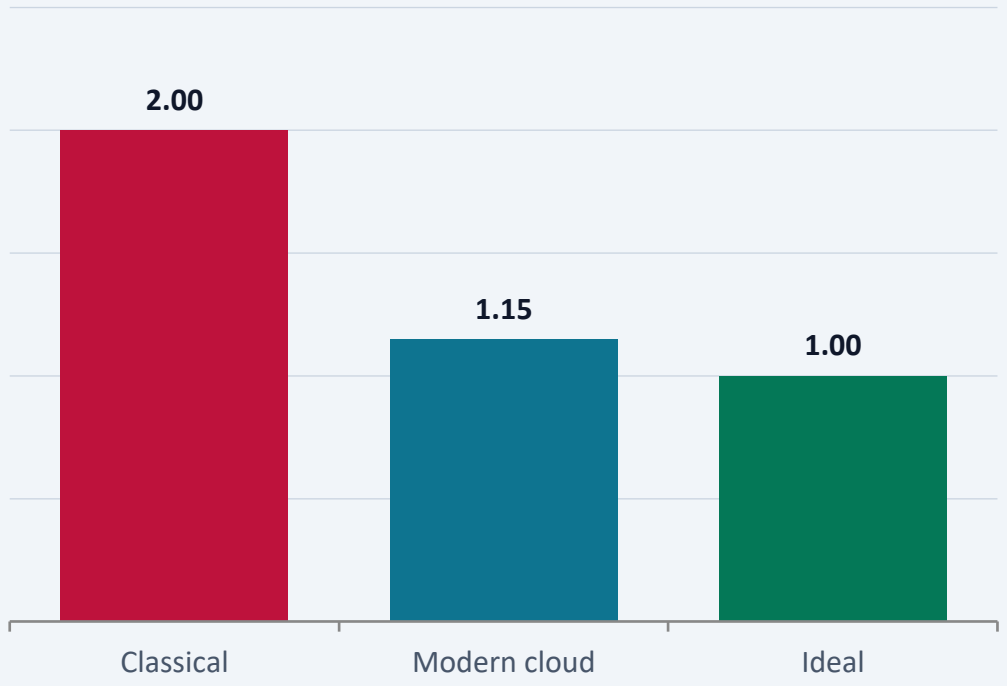
Example: When thousands of GPUs train a large AI model they exchange gradients after every step; RDMA-capable fabrics keep this synchronisation from becoming the bottleneck.

Cooling Infrastructure & the PUE Metric

Power Usage Effectiveness (PUE) is total facility energy divided by the energy used by IT equipment; the ideal value is 1.0. Classical facilities run at about 2.0, meaning cooling and losses use as much power as the servers, while modern cloud data centres reach 1.1 to 1.2. Part of the gain comes from replacing traditional room air handlers (CRAH) with direct-to-chip liquid cooling, where cold plates sit directly on high-power CPUs and GPUs.

Key points

- PUE = total facility power ÷ IT power
- Classical about 2.0; modern cloud 1.1 to 1.2
- CRAH air cooling moves toward liquid cooling
- Cold plates sit directly on CPUs and GPUs

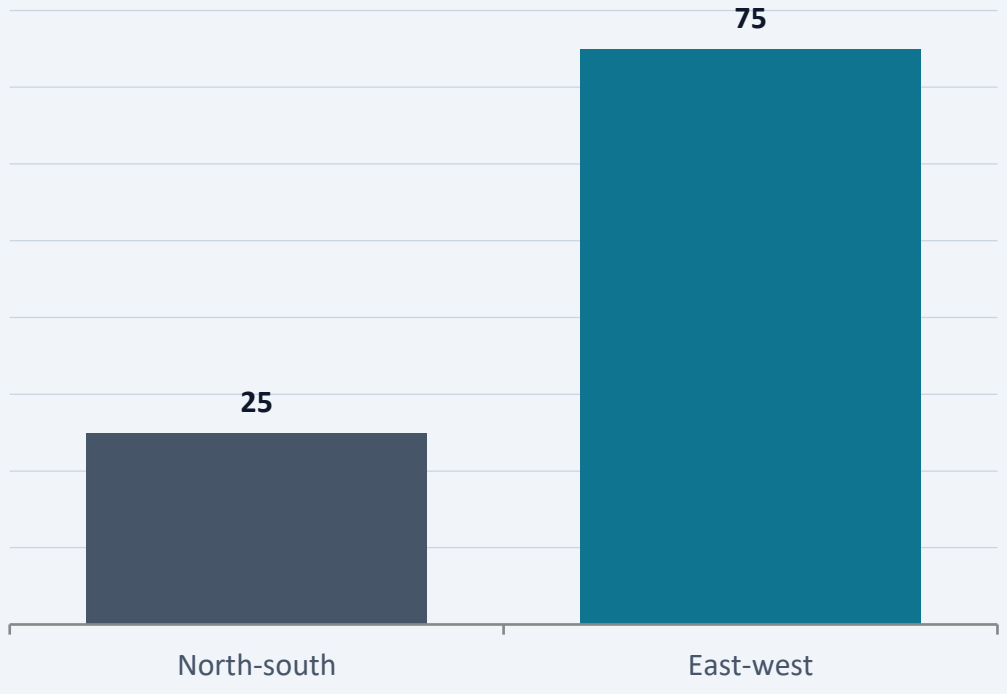


PUE of classical and modern cloud facilities (lower is better)

Example: At a PUE of 2.0 a 1 MW server load needs 2 MW from the grid; at 1.15 it needs only 1.15 MW, saving 0.85 MW continuously.

Nature of Data Centre Traffic

Traffic in a cloud data centre falls into two groups. North-south traffic is user interaction over the WAN or Internet and makes up roughly 20 to 30 per cent. East-west traffic between servers makes up 70 to 80 per cent and includes microservice calls, database replication, MapReduce shuffles and live VM migration. This dominance of east-west traffic drives data centre design away from the 3-tier tree.



Share of traffic in % (midpoints of the 20-30% and 70-80% ranges)

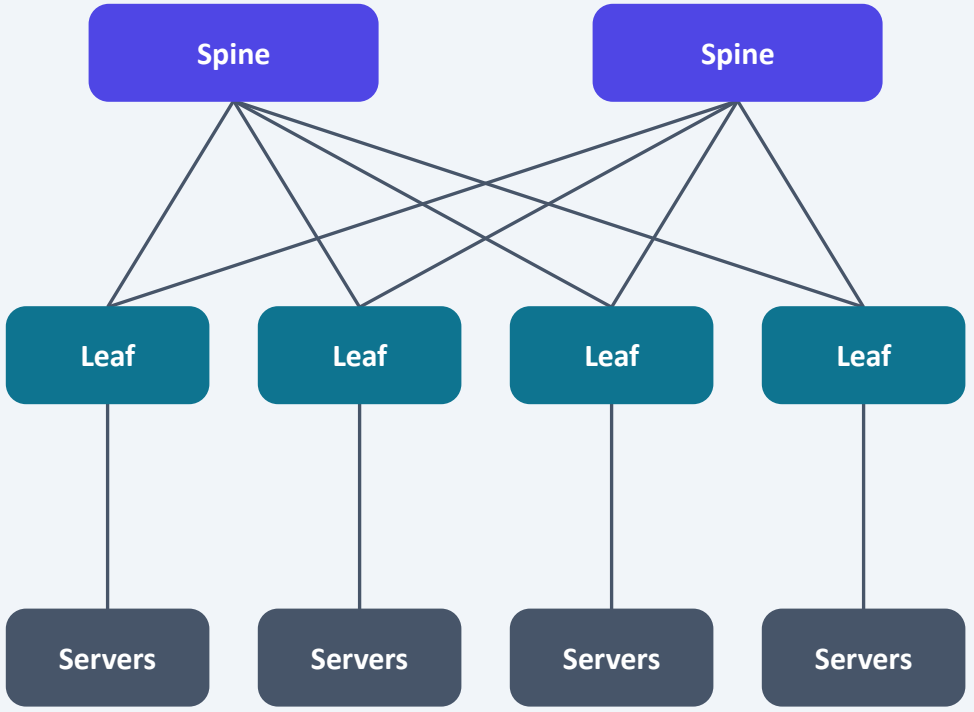
Key points

- North-south: 20-30%, users to data centre
- East-west: 70-80%, server to server
- RPCs, replication, shuffles, VM migration
- Design must favour server-to-server paths

Example: One web search can trigger hundreds of internal calls to index, ranking and advert services; the user sees one response, but the data centre carries hundreds of east-west messages.

Leaf-Spine (Clos) Fabric

To serve heavy east-west traffic, data centres replace the 3-tier tree with a leaf-spine (Clos) fabric. Every leaf switch, which sits at the top of a rack, connects to every spine switch. Any two servers in different racks are therefore the same number of hops apart, so latency is equal and predictable and many equal-cost paths share the load. Capacity grows by adding spine switches, and the fabric can be built non-blocking.



Leaf-spine: full mesh between leaf and spine layers

Key points

- Every leaf connects to every spine
- Same hop count between any two racks
- Many equal-cost paths share the load
- Scale out by adding leaf or spine switches

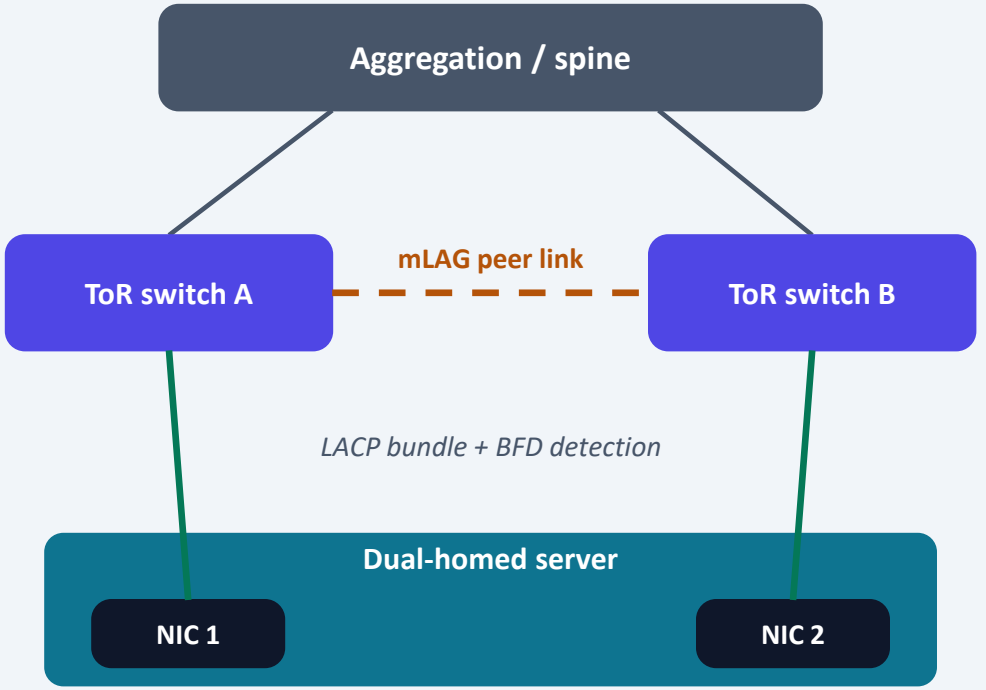
Example: Large web-scale operators build data centres as leaf-spine or multi-stage Clos fabrics from identical commodity switches, which also makes replacing a failed switch routine.

Availability: High-Reliability Mechanisms

Data centre networks must keep running when a cable, port or switch fails. Servers are dual-homed: two network cards connect to two different ToR switches, and the switch pair acts as one logical device through multi-chassis link aggregation (mLAG). LACP bundles physical links into a single logical link so traffic shifts instantly if one member fails, while BFD detects broken paths in milliseconds so routing can reconverge in under a second.

Key points

- Dual NICs to two separate ToR switches
- mLAG makes the switch pair act as one
- LACP (802.3ad) bundles links; instant failover
- BFD detects link breaks in milliseconds



Dual-homed server with mLAG switch pair

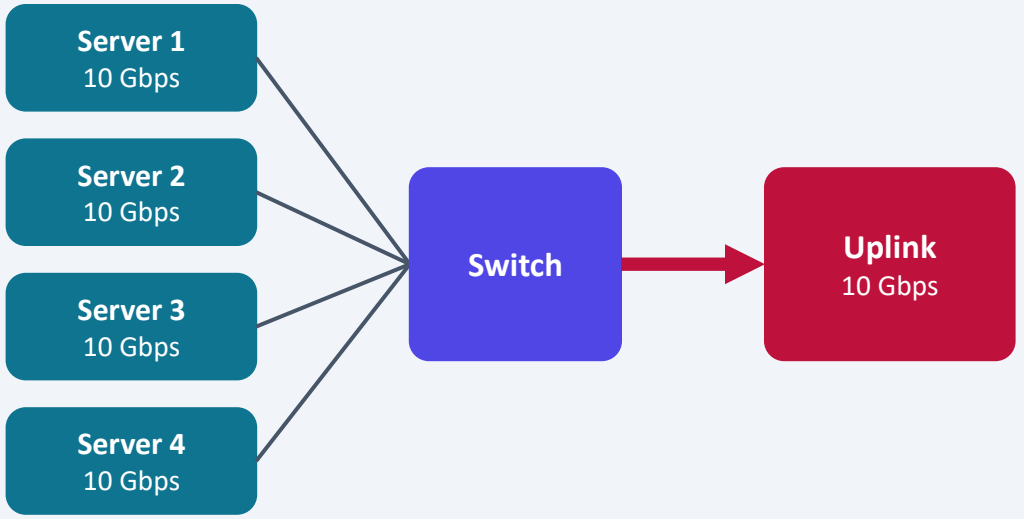
Example: If a technician accidentally unplugs one ToR switch, the server's traffic continues over its second NIC and the application sees no outage.

Poor Performance Factor: Oversubscription

Oversubscription is the ratio between the bandwidth servers can send into a switch and the bandwidth of that switch's uplink. At 4:1, 40 Gbps of server capacity competes for 10 Gbps of uplink. This is economical because servers rarely transmit at full speed together, but when many send bursts at once the uplink queue overflows and packets are dropped. Designers choose the ratio by balancing cost against workload burstiness.

Key points

- Ratio = server downlink : switch uplink
- 4:1 means 40 Gbps competes for 10 Gbps
- Bursts cause queue overflow and packet drops
- Lower ratios cost more but perform better



4:1 oversubscription: four 10G server links share one 10G uplink

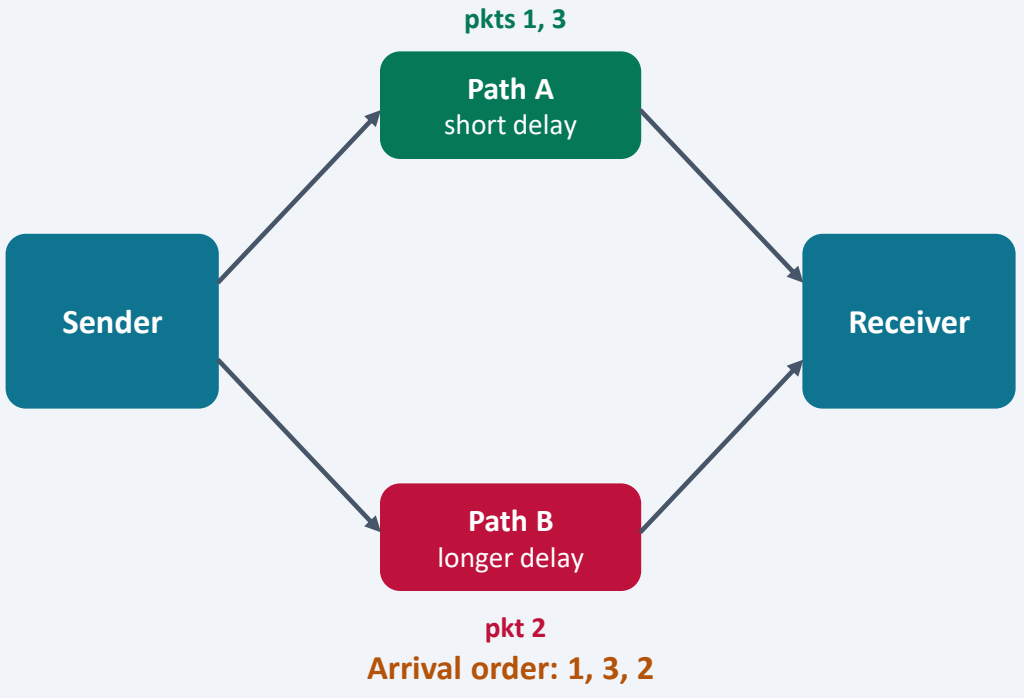
Example: A rack of 40 servers at 10 Gbps (400 Gbps) with a single 100 Gbps uplink is 4:1 oversubscribed; during a Hadoop shuffle all servers send at once and the uplink becomes the bottleneck.

Poor Performance Factor: Jitter & Packet Reordering

Equal-Cost Multi-Path (ECMP) routing spreads traffic over several paths to use the available capacity. If packets of one flow are sent over different paths, each path may have a different queueing delay, so packets arrive out of order and with variable delay (jitter). Applications and TCP then waste time reordering or retransmitting, which raises latency even though nothing in the network is broken.

Key points

- ECMP spreads load across equal-cost paths
- Splitting a single flow mixes path delays
- Different delays reorder packets
- Result: jitter and higher application latency

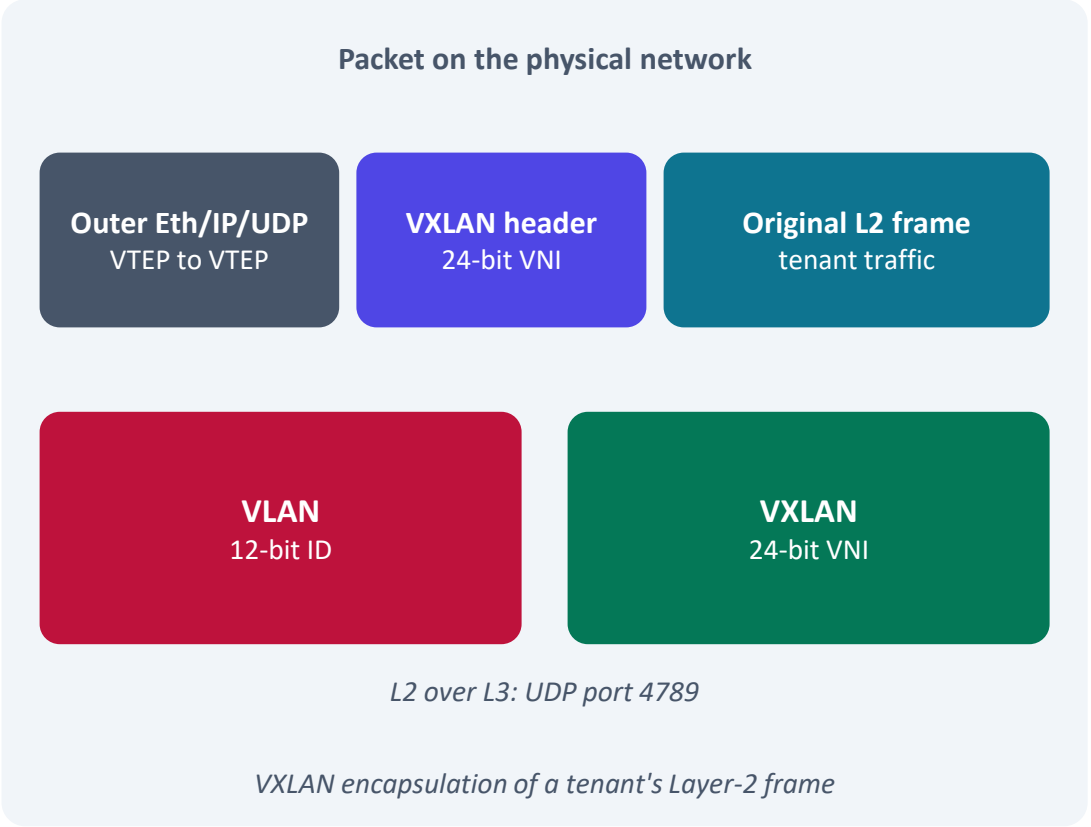


A flow split over two paths arrives out of order

Example: In a video call or a trading feed, packets that arrive as 1, 3, 2 instead of 1, 2, 3 must be buffered and reordered, adding delay that users notice.

Multi-Tenant Security: VXLAN Network Overlays

A cloud data centre hosts many tenants whose networks must be isolated yet run on shared hardware. Traditional VLANs offer only 4,096 IDs, far too few. VXLAN encapsulates a Layer-2 Ethernet frame inside a Layer-3 UDP packet and tags it with a 24-bit network identifier, giving about 16 million isolated networks. Because the overlay rides on IP, tenant networks can span racks and even data centres.



VXLAN encapsulation of a tenant's Layer-2 frame

Key points

- VLAN ID is 12 bits: only 4,096 networks
- VXLAN VNI is 24 bits: about 16 million
- L2 frame wrapped in an L3 UDP packet
- Tenant networks can span racks and sites

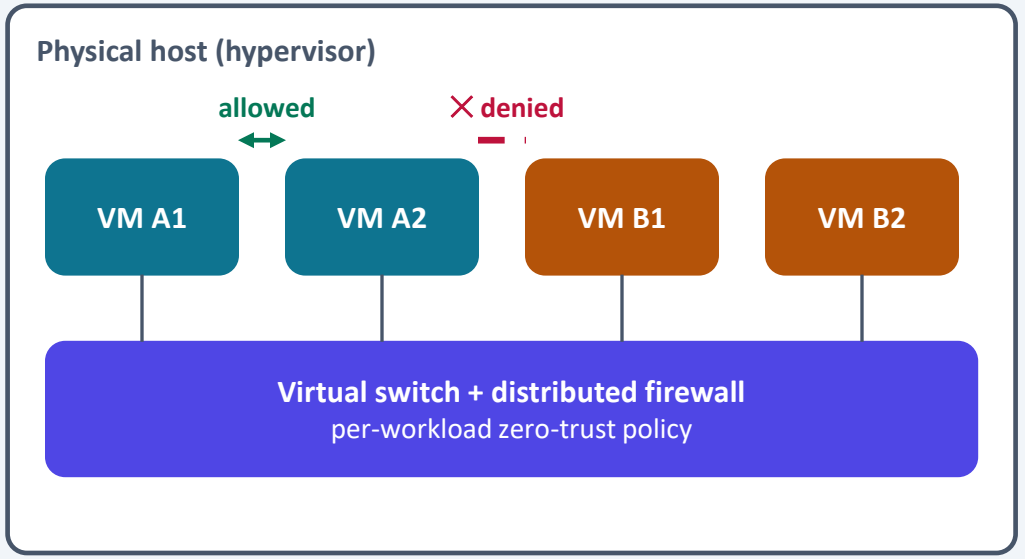
Example: Two companies can both use the private range 10.0.0.0/24 in the same cloud; VXLAN keeps their traffic in separate virtual networks, so there is no address clash and no leakage.

Multi-Tenant Security: Micro-segmentation

Perimeter firewalls protect traffic entering the data centre but cannot see traffic between workloads on the same host. Micro-segmentation applies zero-trust firewalls directly in the hypervisor's virtual switch, so every virtual machine has its own policy. Traffic between workloads is denied by default and allowed only where a rule exists, which limits how far an attacker can move sideways after compromising one VM.

Key points

- Firewall enforced at the hypervisor vSwitch
- Policy follows each workload
- Deny by default; allow by explicit rule
- Limits lateral movement after a breach



Deny by default · allow by explicit rule

Workload-level firewalling inside the hypervisor virtual switch

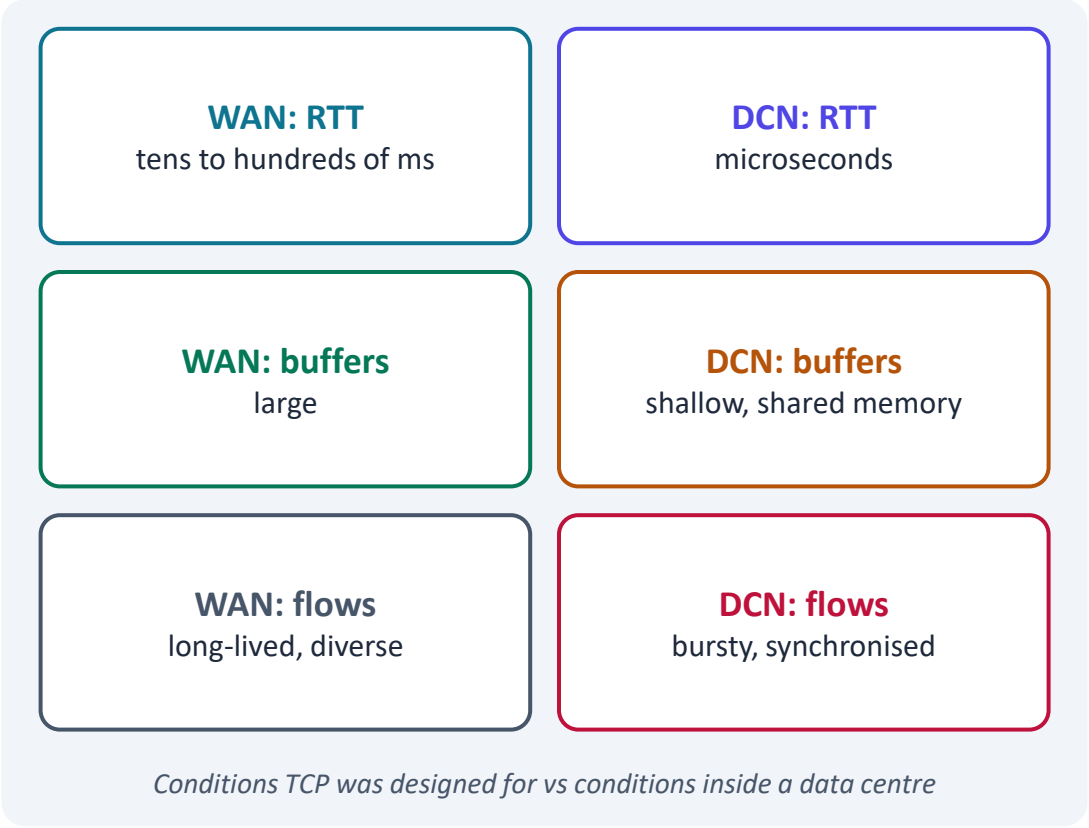
Example: A compromised test VM on the same physical host as a payments VM cannot even reach it, because no rule allows traffic between those two tenants' workloads.

Transport Layer Issues: Why TCP Struggles in DCNs

Traditional TCP congestion control, such as Reno and CUBIC, was designed for wide-area networks with long round-trip times and large switch buffers. A data centre is the opposite: round-trip times are measured in microseconds and switches have shallow buffers. Under heavy burst traffic the standard mechanisms react too late or too harshly, causing the impairments studied in this module: incast, outcast, queue build-up, buffer pressure and pseudocongestion.

Key points

- TCP Reno and CUBIC assume WAN conditions
- DCN: microsecond RTT, shallow buffers
- Bursts overwhelm buffers faster than TCP reacts
- Five key impairments follow



Conditions TCP was designed for vs conditions inside a data centre

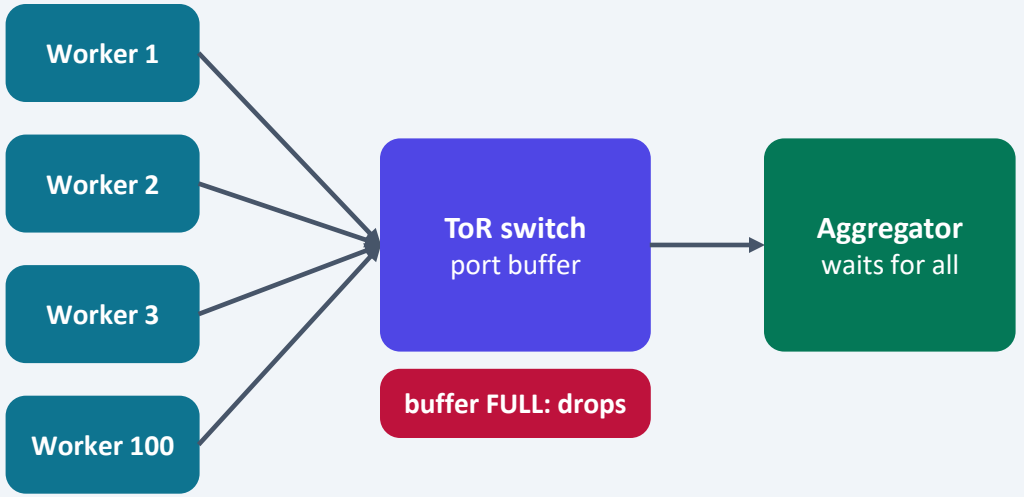
Example: A 200 ms timeout is barely noticeable on a 100 ms WAN path, but when a data centre request normally completes in under a millisecond it is a delay hundreds of times longer.

TCP Incast Collapse

Incast occurs in barrier-synchronised parallel applications such as MapReduce or distributed search. An aggregator sends a request to about 100 workers and waits for all replies before continuing. All workers answer at once, the burst overwhelms the buffer of the ToR switch port in front of the aggregator, and many packets are dropped. Senders then wait for a Retransmission Timeout, at least 200 ms in standard Linux, so throughput can fall by up to 90%.

Key points

- Aggregator queries 100 workers at once
- Synchronised replies overflow the ToR buffer
- Drops force 200 ms retransmission timeouts
- Remedy: microsecond RTO and DCTCP



Many workers reply at once and overflow the switch buffer

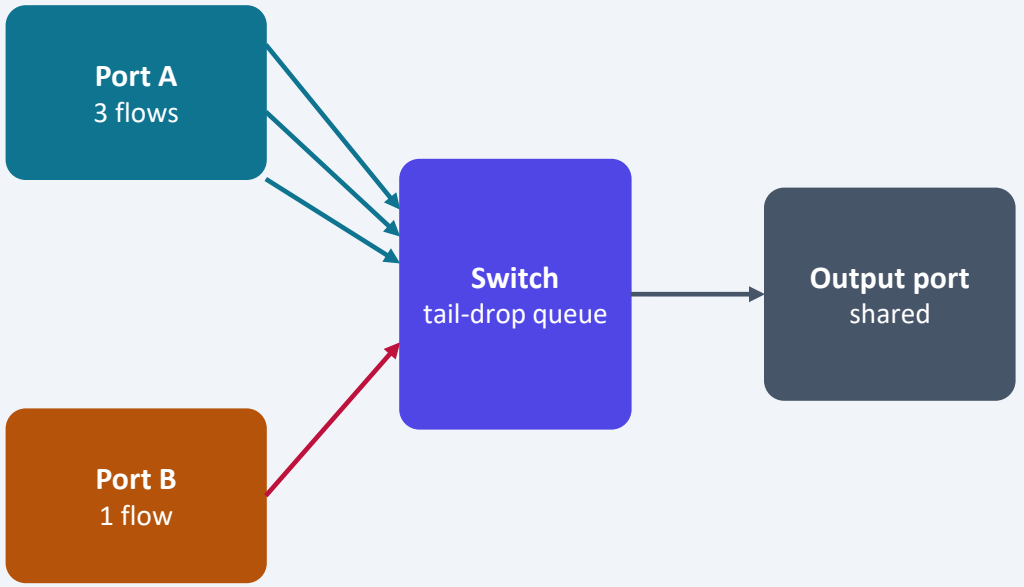
Example: A search front-end waits for 100 index servers. If just one reply is lost and must wait 200 ms for a timeout, the whole query is delayed, even though the others answered within a millisecond.

TCP Outcast: Port Unfairness

Outcast arises when traffic from two input ports of a switch leaves through the same output port, but one input port carries many flows while the other carries only one. With tail-drop queues and shallow buffers, packets from the two ports compete for the last free buffer slots, and the lone flow loses a disproportionate share of its packets. Its throughput collapses, giving unfair sharing caused purely by position in the topology.

Key points

- Many flows on one input port, one on another
- Both leave through the same output port
- Tail drop punishes the single-flow port
- Remedy: RED and weighted fair queuing



Tail drop starves the single flow arriving on port B

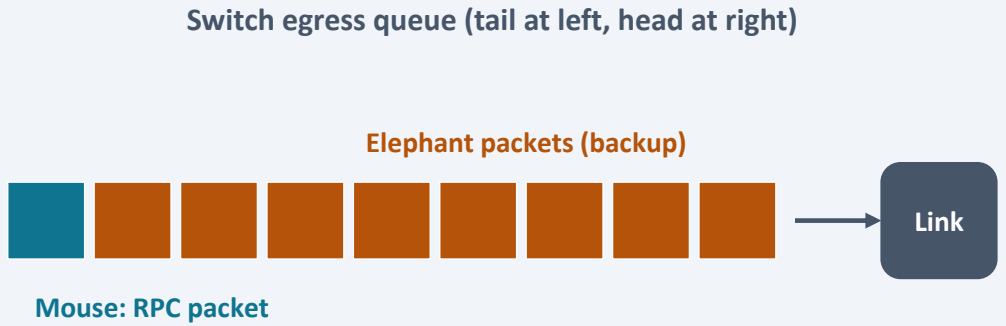
Example: A node receives three flows through one switch port and one flow through another; the lone flow ends up with far less than its fair quarter of the receiver's bandwidth.

Queue Build-up & Latency Spikes

Data centre traffic mixes long-lived, bandwidth-hungry "elephant" flows, such as backups and data migrations, with short, latency-sensitive "mice" flows, such as key-value lookups. Elephants keep egress queues almost permanently full. A small RPC packet arriving at that switch must wait behind the queued elephant packets, so its latency jumps even though the link is not overloaded on average. These tail-latency spikes break service-level agreements.

Key points

- Elephants: long, bulk, throughput-hungry flows
- Mice: short, latency-sensitive requests
- Full queues delay every mouse packet
- Remedy: ECN marking to shrink elephant windows



A mouse packet queued behind elephant packets

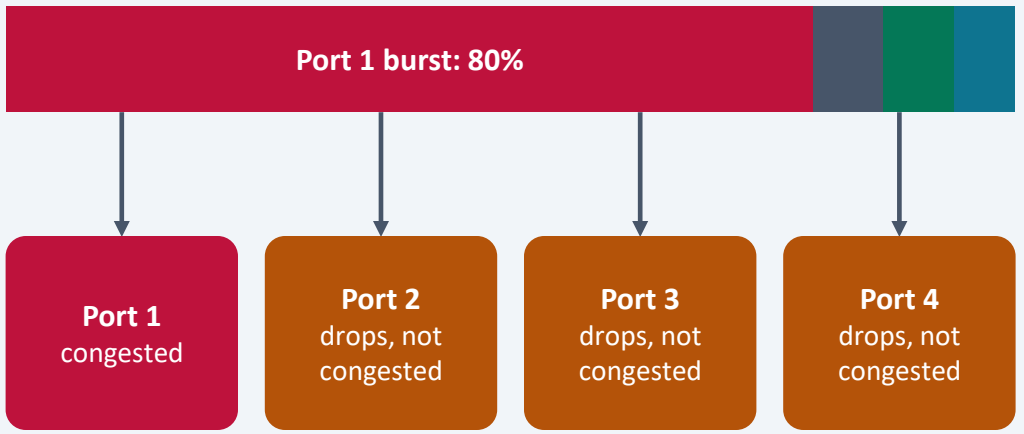
Example: A nightly database backup shares a link with an online shop's cart lookups; during the backup each lookup waits behind megabytes of queued data and the shop feels slow.

Buffer Pressure & Shared Memory Exhaustion

Modern switches use one shared memory pool that is dynamically allocated across all ports, which uses memory efficiently. The weakness is that a heavy burst on one congested port can grab most of the pool. Other ports, which are not congested at all, then find no buffer space and drop their packets too, so a problem on one port harms unrelated traffic across the switch. Dynamic thresholding limits each port's share according to the memory still free.

Key points

- Switch memory is shared among all ports
- One bursting port can drain the whole pool
- Unrelated, uncongested ports then drop packets
- Remedy: dynamic thresholding per port



Unrelated ports lose packets when the pool is exhausted

One bursting port exhausts the shared buffer pool

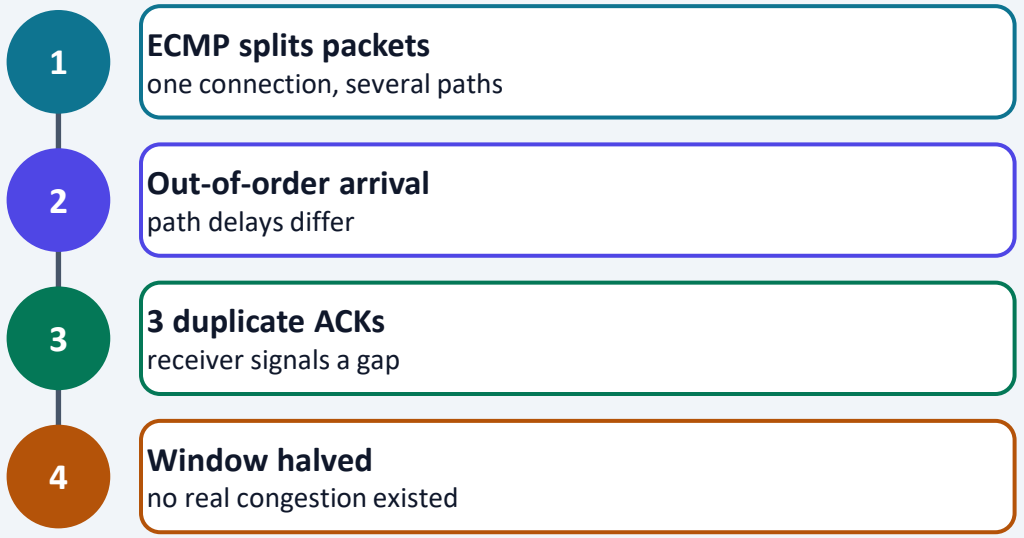
Example: A backup burst on port 1 consumes the shared buffer, so latency-sensitive traffic on ports 2 to 4, carrying only light load, suddenly loses packets.

Pseudocongestion Effect

ECMP can split packets of a single TCP connection over several physical paths. If the paths have different delays, packets arrive out of order. When the receiver sees a gap it sends duplicate ACKs, and after three of them the sender assumes congestion and halves its congestion window. No link was actually congested, so throughput is cut for nothing. Flowlet switching, MP-TCP and QUIC avoid this.

Key points

- ECMP splits one connection across paths
- Out-of-order arrival triggers 3 duplicate ACKs
- Sender halves its window without congestion
- Remedy: flowlet switching, MP-TCP, QUIC



How reordering is mistaken for congestion

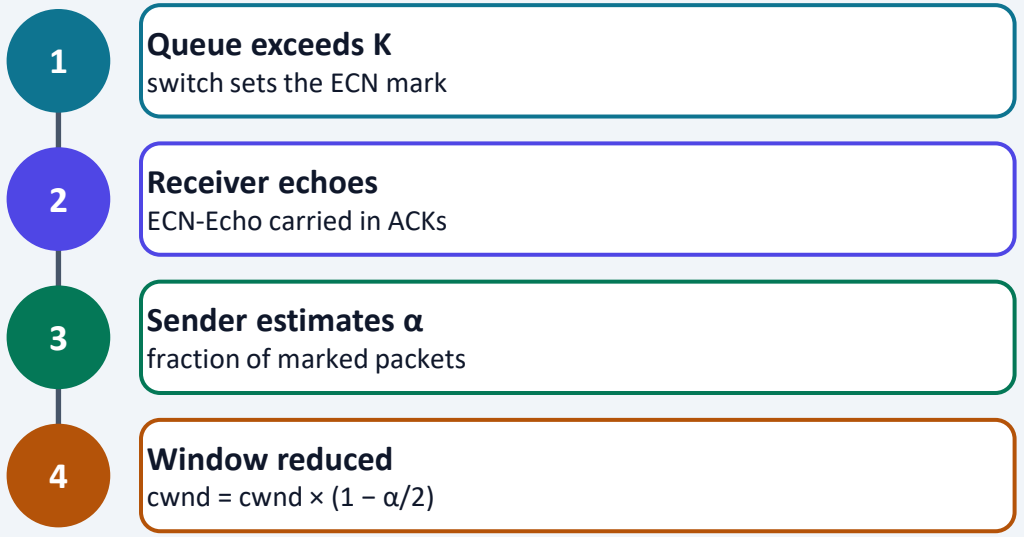
Example: A file transfer over four equal-cost paths runs at only a fraction of line rate because small path-delay differences keep tricking the sender into slowing down.

Remedy Spotlight: ECN and DCTCP

Data Center TCP (DCTCP) keeps queues short using Explicit Congestion Notification. Switches mark packets with a congestion bit as soon as the queue exceeds a small threshold, long before it overflows. The receiver echoes the marks, the sender measures the fraction of marked packets and reduces its window in proportion, instead of always halving it. Elephants back off gently and early, leaving queue space for mice and avoiding drops.

Key points

- Switch marks packets above a queue threshold
- Sender tracks the fraction of marked packets
- Window cut in proportion, not always halved
- Short queues: low latency, high throughput



DCTCP control loop based on ECN marking

Example: With DCTCP a backup flow settles at a small, stable queue length, so an interactive lookup crossing the same switch finds a short queue instead of a full one.

Transport Impairments and Their Remedies

Every impairment below has the same root: microsecond round-trip times and shallow, shared switch buffers combined with bursty, synchronised traffic. The remedies either keep queues short, share buffers fairly, or stop TCP misreading reordering as loss.

Problem	Root cause	Remedy
Incast	Synchronised replies overflow a ToR buffer; 200 ms RTO	Microsecond RTO; DCTCP
Outcast	Unequal flows per input port with tail-drop queues	RED; weighted fair queuing
Queue build-up	Elephant flows keep queues full; mice wait behind them	ECN marking; DCTCP
Buffer pressure	One bursting port consumes the shared memory pool	Dynamic thresholding
Pseudocongestion	ECMP reordering triggers 3 duplicate ACKs	Flowlet switching; MP-TCP; QUIC

Quick Revision Map: Key Terms

3-tier tree
Core, aggregation and access layers

East-west
Server-to-server traffic; 70-80% of DC traffic

STP
Blocks redundant L2 links; wastes up to 50%

SDN / NFV
Separate control and data planes; virtual functions

ToR vs EoR
Switch per rack vs one switch per row

PUE
Total facility power ÷ IT power; ideal is 1.0

Leaf-spine
Every leaf links to every spine; equal latency

LACP / BFD
Link bundling and millisecond failure detection

Oversubscription
Downlink : uplink ratio, e.g. 4:1

VXLAN
L2 over UDP; 24-bit ID gives 16 million networks

Incast
Synchronised replies overflow buffers; 200 ms RTO

DCTCP / ECN
Early marking; window cut in proportion

Likely Exam Questions

1. Compare the 3-tier tree with a leaf-spine fabric. Why does east-west traffic matter?

2. Explain how STP wastes link capacity in a data centre.

3. Describe SDN and NFV and their role in cloud data centres.

4. Explain hot/cold aisle containment and compare ToR with EoR cabling.

5. Differentiate SAN, NAS and SDS. Why are RDMA fabrics used for AI clusters?

6. Define PUE. How do modern facilities lower it?

7. Explain high availability using dual-homing, mLAG, LACP and BFD.

8. What is oversubscription? Explain a 4:1 ratio with an example.

9. Explain how VXLAN and micro-segmentation support multi-tenancy.

10. Explain incast, outcast, queue build-up, buffer pressure and pseudocongestion with remedies.

Answer tip: define the term, sketch the labelled diagram, list 3 to 4 key points, and close with an example or remedy. For Module 3, always state the scenario, the mechanism and the remedy.

References & Further Reading

1 **Essentials of Cloud Computing**
K. Chandrasekaran, CRC Press (Taylor & Francis), 2015: background on virtualisation, cloud data centres and service delivery.

2 **A Scalable, Commodity Data Center Network Architecture**
M. Al-Fares, A. Loukissas, A. Vahdat, SIGCOMM 2008: fat-tree / Clos-style fabrics.

3 **Data Center TCP (DCTCP)**
M. Alizadeh et al., SIGCOMM 2010: ECN-based congestion control for data centres.

4 **The TCP Outcast Problem**
P. Prakash et al., NSDI 2012: port unfairness with tail-drop switches. For incast see Phanishayee et al., FAST 2008.

5 **VXLAN**
IETF RFC 7348: Virtual eXtensible Local Area Network overlay framework.